

Cisco IOS & Networking Mastery Notes

From Zero to Advanced CLI Configuration

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1. Networking Foundations

1.1 The OSI Model (7 Layers)

Layer	Name	PDU	Function	Example Protocols
7	Application	Data	User-facing services	HTTP, FTP, DNS, SMTP
6	Presentation	Data	Encryption, compression, formatting	SSL/TLS, JPEG
5	Session	Data	Session establishment/teardown	NetBIOS, RPC
4	Transport	Segment	End-to-end delivery, reliability	TCP, UDP
3	Network	Packet	Logical addressing, routing	IP, ICMP, OSPF
2	Data Link	Frame	Physical addressing, framing	Ethernet, PPP, VLAN (802.1Q)
1	Physical	Bit	Electrical/optical transmission	Cabling, connectors, signaling

Mnemonic: **All People Seem To Need Data Processing.**

1.2 TCP/IP Model (4/5 Layers)

TCP/IP Layer	Maps to OSI
Application	5,6,7

TCP/IP Layer	Maps to OSI
Transport	4
Internet	3
Network Access (Link + Physical)	1,2

1.3 Encapsulation / De-encapsulation

Data → Segment (L4, adds port#) → Packet (L3, adds IP) → Frame (L2, adds MAC) → Bits (L1). De-encapsulation is the reverse on the receiving device. Every device that forwards a frame at L2 (switch) only inspects MAC; every device that routes at L3 (router) strips and rebuilds the L2 frame each hop while the L3 packet's source/destination IP stays the same end-to-end (unless NAT).

1.4 TCP vs UDP

- **TCP**: connection-oriented, 3-way handshake (SYN, SYN-ACK, ACK), sequencing, acknowledgment, flow control (windowing), retransmission, teardown (FIN/ACK x4 or RST). Used by HTTP, FTP, SSH, Telnet, SMTP.
- **UDP**: connectionless, no handshake, no guaranteed delivery, low overhead. Used by DNS, DHCP, TFTP, VoIP (RTP), SNMP, syslog.

1.5 IPv4 Addressing

- 32-bit address, 4 octets, dotted decimal (e.g., 192.168.1.1).
- Classes (historical, no longer used for routing decisions but still relevant for defaults):
 - Class A: 0.0.0.0–127.255.255.255 (default /8)
 - Class B: 128.0.0.0–191.255.255.255 (default /16)
 - Class C: 192.0.0.0–223.255.255.255 (default /24)
 - Class D: 224.0.0.0–239.255.255.255 (Multicast)
 - Class E: 240.0.0.0–255.255.255.255 (Experimental)
- Private ranges (RFC1918): 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16.
- Loopback: 127.0.0.0/8. APIPA: 169.254.0.0/16.

1.6 Subnetting (CIDR)

- CIDR notation: /prefix length = number of network bits.
- Subnet mask examples: /24=255.255.255.0, /25=255.255.255.128, /26=.192, /27=.224, /28=.240, /29=.248, /30=.252, /32=.255.255.255.255 (host route).
- **Magic number method**: 256 – (last non-255 octet of mask) = block size.

- Example: /27 → mask 255.255.255.224 → block size = 256-224 = 32 → subnets: .0, .32, .64, .96, .128, .160, .192, .224.
- Usable hosts per subnet = $2^{(32-\text{prefix})} - 2$ (subtract network & broadcast address). /30 = 2 usable hosts, ideal for point-to-point WAN links.
- VLSM (Variable Length Subnet Masking): subnetting a subnet — assign the largest host requirement first, smallest last, to conserve address space.
- Route summarization (supernetting): combine multiple contiguous subnets into one larger CIDR block advertised upstream, reduces routing table size.

1.7 Switching Concepts

- Switches operate at Layer 2, forward frames using the **MAC address table (CAM table)** learned dynamically from source MAC + ingress port.
- Unknown unicast, broadcast, and multicast frames are **flooded** out all ports except the ingress port (within the same VLAN/broadcast domain).
- **Collision domain**: segment where frames can collide (each switch port = its own collision domain in full duplex, eliminating collisions).
- **Broadcast domain**: extent to which broadcasts propagate; bounded by routers or VLAN boundaries.

1.8 Routing Concepts

- Routers operate at Layer 3, forward packets based on the **routing table**, using the longest prefix match.
- Administrative Distance (AD) — trustworthiness of a route source (lower = more preferred): | Source | AD | |---|---| | Connected | 0 | | Static | 1 | | EIGRP summary | 5 | | External BGP | 20 | | Internal EIGRP | 90 | | OSPF | 110 | | IS-IS | 115 | | RIP | 120 | | External EIGRP | 170 | | Internal BGP | 200 | | Unknown/Unreachable | 255 |
- **Metric**: cost value used to select best path within the *same* protocol (hop count for RIP, bandwidth/delay composite for EIGRP, cost=100Mbps/BW for OSPF).

2. Cisco IOS Fundamentals

2.1 What is IOS?

Cisco **Internetwork Operating System (IOS)** is the CLI-driven OS running on Cisco routers/switches. Newer platforms may run **IOS-XE** (Linux-based, modular, supports containers) but the classic CLI syntax you learn here applies almost identically.

2.2 Accessing the Device

- **Console port:** direct physical/serial (RJ45/USB) connection, required for initial setup, out-of-band, works even with no network config (default 9600 8N1 baud).
- **Telnet:** remote, unencrypted (avoid in production).
- **SSH:** remote, encrypted — preferred.
- **AUX port:** legacy modem access.

2.3 CLI Modes (Hierarchy)

User EXEC	Router>	limited, show-only, no config
Privileged EXEC	Router#	full show/debug/clear, "enable"
Global Config	Router(config)#	"configure terminal"
Interface Config	Router(config-if)#	"interface gi0/0"
Line Config	Router(config-line)#	"line vty 0 4" / "line console 0"
Router Config	Router(config-router)#	"router ospf 1"
VLAN Config	Router(config-vlan)#	"vlan 10"
Sub-interface	Router(config-subif)#	"interface gi0/0.10"

Move up one level: **exit**. Jump straight to privileged exec from anywhere: **end** or **Ctrl+Z**.

2.4 Getting Help & Navigation

Command/Key	Function
?	Context-sensitive help (list commands or arguments)
Tab	Auto-complete a partial command
Ctrl+A	Move cursor to beginning of line
Ctrl+E	Move cursor to end of line
Ctrl+R	Redisplay/refresh current line

Command/Key	Function
Ctrl+W	Erase a word
Ctrl+U	Erase entire line
Ctrl+Shift+6 then X	Break/abort an ongoing process (e.g. stuck ping/traceroute)
Ctrl+Z	Exit to privileged EXEC from any config mode
Up/Down arrow or Ctrl+P/Ctrl+N	Cycle through command history
show history	List command history buffer
terminal history size <n>	Increase history buffer (default 10)

2.5 Command Abbreviation & Do-command

- IOS allows shortest-unambiguous abbreviation: `conf t = configure terminal, sh run = show running-config, int gi0/0 = interface gigabitEthernet0/0.`
- From any config sub-mode, prefix `do` to run an EXEC-level command without leaving config mode: `Router(config-if)#do show ip interface brief.`

2.6 Configuration Files

- **running-config**: active configuration in RAM; lost on reload unless saved.
- **startup-config**: saved configuration in NVRAM; loaded on boot.

show running-config (or: show run)

show startup-config (or: show start)

copy running-config startup-config (or: write memory / wr)

copy startup-config running-config

write erase ! erases startup-config

erase startup-config

reload ! reboots device

2.7 show and clear — the Backbone Commands

show version ! IOS version, uptime, hardware, config register

show running-config

show startup-config

show interfaces [int] ! detailed L1/L2/L3 interface stats

show ip interface brief ! quick status/IP summary

show ip route ! routing table

show mac address-table ! CAM table (switches)

show vlan brief

show cdp neighbors [detail]

show lldp neighbors [detail]

show inventory

show processes cpu

show memory statistics

show flash:

show logging

show arp

show users

show sessions

2.8 Basic Global Settings

enable

configure terminal

hostname R1

no ip domain-lookup ! stop CLI from trying to DNS-resolve mistyped commands (huge QoL)

banner motd #Unauthorized access is prohibited#

banner login #Authorized users only#

enable secret Cisco123! ! MD5/SHA-256 hashed enable password (preferred over 'enable password')

service password-encryption ! weakly (type 7) encrypts other plaintext passwords in config

line console 0

password ConsolePass

login

exec-timeout 10 0 ! 10 min timeout, avoid idle console sessions

logging synchronous ! prevents log messages interrupting typed commands

exit

line vty 0 4

password VtyPass

login

transport input ssh ! restrict remote access to SSH only

exit

end

copy run start

3. Initial Device Configuration (Full Walkthrough)

3.1 SSH Setup (Detailed)

```
hostname R1
```

```
ip domain-name lab.local      ! required before generating RSA keys
```

```
crypto key generate rsa modulus 2048 ! generates the key pair, enables SSH
```

```
ip ssh version 2
```

```
username admin privilege 15 secret Str0ngP@ss
```

```
line vty 0 4
```

```
transport input ssh
```

```
login local                  ! authenticate against local username DB
```

```
exit
```

Verify: `show ip ssh`, `show crypto key mypubkey rsa`.

3.2 Local User Database & Privilege Levels

```
username netadmin privilege 15 secret Cisco@123
```

```
privilege exec level 5 show running-config ! custom privilege mapping (advanced/rare)
```

Privilege levels: 0 (minimal: logout, enable, disable, help), 1 (user exec default), 15 (full privileged exec). Levels 2–14 are custom and rarely used without `privilege` command mappings.

3.3 Banners

```
banner motd ^C
```

```
*****
```

Authorized Access Only

^C

Delimiting character (e.g. ^C or #) must not appear inside the banner text itself.

3.4 Saving, Backing Up, and Restoring Configs

copy running-config startup-config

copy running-config tftp:

copy tftp: running-config

copy running-config flash:backup-config

copy startup-config scp:

Best practice: always back up config before major changes; use a TFTP/SCP server or a version-controlled text export.

3.5 Clock, NTP, and Time Zone

clock set 14:30:00 1 September 2026

clock timezone IST 5 30

ntp server 192.168.1.1

ntp master 3 ! act as an NTP source (usually only on a hub device)

show clock

show ntp status

show ntp associations

3.6 CDP & LLDP (Neighbor Discovery)

cdp run ! global enable (default on)

no cdp run ! disable globally for security on edge-facing devices

interface gi0/1

no cdp enable ! disable per interface

lldp run

show cdp neighbors detail

show lldp neighbors detail

CDP is Cisco-proprietary L2 discovery; LLDP is the vendor-neutral IEEE 802.1AB standard — use LLDP in mixed-vendor environments.

4. Interfaces & IP Addressing

4.1 Interface Naming

Examples: `FastEthernet0/1`, `GigabitEthernet0/0`, `TenGigabitEthernet1/0/1`, `Serial0/0/0`, `Loopback0`, `Vlan10` (SVI), `Port-channel1`, `Tunnel0`.

4.2 Basic Interface Configuration

interface gigabitEthernet0/0

description Link-to-Core-SW

ip address 192.168.10.1 255.255.255.0

no shutdown ! interfaces are administratively DOWN by default on routers

duplex full

speed 1000

exit

Verify:

show ip interface brief

show interfaces gi0/0

show interfaces status ! switches - shows connected/notconnect/err-disabled

show controllers serial 0/0/0 ! shows DCE/DTE, needed for clock rate

On a serial link, the **DCE end** must set clocking: `clock rate 64000` (bps) inside interface config.

4.3 Loopback Interfaces

interface loopback0

ip address 10.10.10.1 255.255.255.255

Always up as long as the device is up (no physical layer) — used for router-IDs (OSPF/BGP), management, and testing.

4.4 IPv4 Address Assignment Methods

ip address 192.168.1.1 255.255.255.0 ! static

ip address dhcp ! interface obtains via DHCP client

ip address negotiated ! PPP/PPPoE negotiated address

4.5 Secondary IP Addresses

interface gi0/0

ip address 192.168.1.1 255.255.255.0

ip address 192.168.2.1 255.255.255.0 secondary

Used when multiple subnets must share one physical segment (legacy technique; prefer sub-interfaces/VLANs where possible).

4.6 Interface Troubleshooting States

show interfaces output	Meaning
up, line protocol up	Fully operational
administratively down, down	shutdown applied
down, down	Physical/Layer1 problem (cable, far end down)
up, line protocol down	L2 issue: encapsulation mismatch, keepalives, no clock rate on serial
err-disabled	Switch disabled port due to violation (e.g., port-security, BPDU guard, storm-control) — fix cause then shutdown / no shutdown to recover, or configure errdisable recovery

errdisable recovery cause psecure-violation

errdisable recovery interval 60

5. Switching: VLANs, Trunking, STP, EtherChannel

5.1 VLANs

VLANs create separate broadcast domains on a shared physical switch fabric.

```
vlan 10
```

```
name SALES
```

```
vlan 20
```

```
name ENGINEERING
```

```
exit
```

```
interface fa0/1
```

```
switchport mode access
```

```
switchport access vlan 10
```

Verify: `show vlan brief`, `show interfaces fa0/1 switchport`.

Voice VLAN (for IP phones sharing a port with a PC):

```
interface fa0/2
```

```
switchport mode access
```

```
switchport access vlan 10
```

```
switchport voice vlan 100
```

5.2 Trunking (802.1Q)

Trunks carry multiple VLANs between switches, tagging frames with a 4-byte 802.1Q header (VLAN ID 1–4094). The **native VLAN** (default VLAN 1) is sent untagged — must match on both ends or you get VLAN mismatch issues/security risk.

```
interface gi0/1
```

```
switchport trunk encapsulation dot1q    ! needed on switches supporting ISL too
```

```
switchport mode trunk
```

```
switchport trunk native vlan 999        ! change from default VLAN1 (best practice)
```

```
switchport trunk allowed vlan 10,20,30  ! prune unnecessary VLANs
```

```
switchport trunk allowed vlan add 40
```

DTP (Dynamic Trunking Protocol): auto-negotiates trunk/access mode (`dynamic auto/dynamic desirable`). Best practice: disable DTP with `switchport nonegotiate` and hard-set modes explicitly to avoid VLAN-hopping attacks.

5.3 VTP (VLAN Trunking Protocol)

Propagates VLAN database across switches in a domain.

```
vtp domain CORP
```

```
vtp mode server      ! server (default) | client | transparent | off
```

```
vtp password Cisco123
```

```
vtp version 2
```

```
show vtp status
```

- **Server:** can create/modify/delete VLANs, propagates changes.
- **Client:** cannot modify VLANs locally, only receives.
- **Transparent:** forwards VTP ads but keeps its own independent local VLAN DB — recommended in modern networks to avoid the infamous "VTP revision number disaster" (a switch with a higher revision number can wipe out the whole domain's VLANs). Mitigate by always resetting revision to 0 before connecting a new switch (**vtp mode transparent** then back, or change domain name twice).

5.4 Spanning Tree Protocol (STP)

Prevents Layer 2 loops in redundant topologies by blocking redundant paths.

- **Root Bridge election:** lowest Bridge ID (Priority + MAC) wins. Default priority 32768.
- **Port Roles:** Root Port (best path to root, one per non-root switch), Designated Port (best path per segment), Blocking/Alternate Port (loop prevention).
- **Port States (classic 802.1D):** Blocking → Listening → Learning → Forwarding (each transition ~15–20s, total ~30–50s convergence — slow!).

```
spanning-tree mode pvst      ! PVST+ (default, per-VLAN)
```

```
spanning-tree mode rapid-pvst  ! RSTP per VLAN — fast convergence (sub-second)
```

```
spanning-tree mode mst        ! Multiple Spanning Tree — maps many VLANs to few instances
```

```
spanning-tree vlan 10 priority 4096 ! manually influence root election (lower=more preferred, increments of 4096)
```

spanning-tree vlan 10 root primary ! auto-sets priority lower than current root

spanning-tree vlan 10 root secondary

STP Enhancements (must-know for real deployments):

interface fa0/1

spanning-tree portfast ! skips listening/learning on access ports (end-hosts only!)

spanning-tree bpduguard enable ! err-disables port if a BPDU is received on a PortFast port (rogue switch protection)

spanning-tree portfast bpduguard default ! global enable for all portfast ports

spanning-tree guard root ! prevents a downstream switch from becoming root (per trunk interface)

spanning-tree loopguard default

Verify: `show spanning-tree`, `show spanning-tree vlan 10`, `show spanning-tree interface fa0/1 detail`.

5.5 EtherChannel (Link Aggregation)

Bundles multiple physical links into one logical link for bandwidth + redundancy; STP treats the bundle as a single link (no blocking).

interface range gi0/1 - 2

channel-group 1 mode active ! LACP (IEEE standard) — active/passive

channel-group 1 mode desirable ! PAgP (Cisco proprietary) — desirable/auto

channel-group 1 mode on ! static, no negotiation protocol (not recommended)

interface port-channel 1

switchport mode trunk

Rule: all bundled physical ports must match — speed, duplex, VLAN/trunk config, STP settings — or bundling fails. Verify: `show etherchannel summary`, `show etherchannel port-channel`. **Load-balancing method:**

port-channel load-balance src-dst-ip

6. Inter-VLAN Routing

6.1 Router-on-a-Stick

One physical router interface, sub-interfaced per VLAN, connected to a trunk port on the switch.

```
interface gi0/0.10
```

```
encapsulation dot1Q 10
```

```
ip address 192.168.10.1 255.255.255.0
```

```
interface gi0/0.20
```

```
encapsulation dot1Q 20
```

```
ip address 192.168.20.1 255.255.255.0
```

```
interface gi0/0.999
```

```
encapsulation dot1Q 999 native    ! native VLAN sub-interface, no tag added
```

Switch side:

```
interface gi0/1
```

```
switchport mode trunk
```

6.2 SVI (Switched Virtual Interface) on a Layer 3 Switch

Preferred in modern designs (faster, no single trunk bottleneck):

```
ip routing    ! must enable L3 routing capability on the switch
```

```
interface vlan 10
```

```
ip address 192.168.10.1 255.255.255.0
```

```
no shutdown
```

```
interface vlan 20
```

```
ip address 192.168.20.1 255.255.255.0
```

6.3 Routed Ports on L3 Switches

```
interface gi0/1
```

```
no switchport          ! converts a switch port into a pure L3 routed port
```

```
ip address 10.0.0.1 255.255.255.252
```

7. Static & Default Routing

7.1 Static Route Syntax

```
ip route <destination-network> <subnet-mask> <next-hop-ip | exit-interface> [AD] [name X]
```

```
ip route 192.168.20.0 255.255.255.0 10.0.0.2
```

```
ip route 192.168.20.0 255.255.255.0 gi0/0/1      ! exit-interface style (only for P2P links)
```

```
ip route 0.0.0.0 0.0.0.0 203.0.113.1            ! default route ("gateway of last resort")
```

```
ip route 192.168.20.0 255.255.255.0 10.0.0.2 200 ! floating static (higher AD = backup route)
```

- Prefer **next-hop IP** style on multi-access (Ethernet) segments to avoid proxy-ARP/recursive lookup issues; exit-interface style is fine (even preferred) only on true point-to-point (serial/PPP) links.
- **Floating static routes** act as backup: configured with a higher AD than the primary dynamic/static route so they're only installed in the RIB when the primary fails.

7.2 Verifying

```
show ip route
```

```
show ip route static
```

```
show ip route 192.168.20.0
```

```
traceroute 192.168.20.5
```

Route table codes: **C**=connected, **S**=static, **O**=OSPF, **D**=EIGRP, **R**=RIP, **B**=BGP, **L**=local (host route /32 auto-created per interface), *****=candidate default, **>**=next hop marker.

8. Dynamic Routing Protocols

8.1 RIP (Routing Information Protocol) — legacy, know it conceptually

Distance-vector, hop-count metric (max 15, 16=unreachable), updates every 30s, slow convergence.

```
router rip
```

```
version 2
```

```
network 192.168.1.0
```

```
no auto-summary
```

```
passive-interface gi0/0      ! suppress RIP updates out user-facing interfaces
```

8.2 EIGRP (Enhanced Interior Gateway Routing Protocol) — Cisco advanced distance-vector (hybrid)

Uses **DUAL (Diffusing Update Algorithm)** for loop-free, fast convergence. Metric = composite of bandwidth + delay (+ optionally load/reliability, rarely used).

```
router eigrp 100              ! classic mode, AS number must match neighbors
```

```
network 192.168.1.0 0.0.0.255
```

```
no auto-summary
```

```
eigrp router-id 1.1.1.1
```

```
passive-interface gi0/0
```

Named mode (modern, recommended):

```
router eigrp NAMED-PROC
```

```
address-family ipv4 unicast autonomous-system 100
```

```
network 192.168.1.0 0.0.0.255
```

```
af-interface gi0/0
```

```
passive-interface
```

```
exit-af-interface
```

```
exit-address-family
```

Key terms: **Successor** (best route, installed in RIB), **Feasible Successor** (backup route, pre-computed, satisfies feasibility condition — instant failover with no recomputation), **Feasibility Condition** (Reported Distance of neighbor < Feasible Distance of current successor). Verify:

```
show ip eigrp neighbors
```

```
show ip eigrp topology
```

```
show ip route eigrp
```

```
show ip protocols
```

Authentication: `key chain`, then under interface: `ip authentication mode eigrp 100 md5 / ip authentication key-chain eigrp 100 CHAIN1.`

8.3 OSPF (Open Shortest Path First) — Link-state, industry standard

Uses **Dijkstra's SPF algorithm**. Metric = cost = reference-bandwidth / interface-bandwidth (default reference 100 Mbps — adjust with **auto-cost reference-bandwidth** on modern high-speed links).

```
router ospf 1
```

```
router-id 1.1.1.1
```

```
network 192.168.1.0 0.0.0.255 area 0
```

```
network 10.0.0.0 0.0.0.3 area 0
```

```
passive-interface gi0/0
```

```
auto-cost reference-bandwidth 10000    ! in Mbps, so links >100M are differentiated
```

Per-interface alternative (modern, especially on L3 switches):

```
interface gi0/0
```

```
ip ospf 1 area 0
```

```
ip ospf cost 20
```

```
ip ospf priority 255    ! influences DR/BDR election, higher wins
```

```
ip ospf hello-interval 5
```

```
ip ospf dead-interval 20
```

OSPF Concepts:

- **Areas:** Area 0 (backbone) is mandatory in multi-area designs; all other areas must connect to Area 0 (directly or via virtual-link).
- **DR/BDR election:** on multi-access (broadcast/NBMA) segments only, to reduce adjacency flooding; elected by highest priority, tie-break highest Router ID. Point-to-point serial links skip DR/BDR.

- **LSA Types:** Type 1 (Router), Type 2 (Network — from DR), Type 3 (Summary/inter-area, from ABR), Type 4 (ASBR Summary), Type 5 (External, from ASBR — redistributed routes), Type 7 (NSSA External, converted to Type 5 at the NSSA ABR).
- **Area types:** Standard, Stub (blocks Type 5, injects default route), Totally Stubby (Cisco-proprietary, blocks Type 3+4+5 too), NSSA (Not-So-Stubby — allows limited external redistribution via Type 7).

area 10 stub

area 10 stub no-summary ! totally stubby (on ABR only)

area 20 nssa

- **Router types:** Internal (all interfaces in one area), Backbone (interface in area 0), ABR — Area Border Router (interfaces in multiple areas), ASBR — Autonomous System Boundary Router (redistributes external routes into OSPF). Verify:

show ip ospf neighbor

show ip ospf interface brief

show ip ospf database

show ip route ospf

show ip protocols

OSPF neighbor states: Down → Attempt → Init → 2-Way → ExStart → Exchange → Loading → **Full** (fully synchronized).

8.4 BGP (Border Gateway Protocol) — Path-vector, Internet-scale routing

Used between Autonomous Systems (eBGP) or within large enterprises (iBGP). Not metric-based like IGPs — uses a **best-path selection algorithm** based on attributes.

router bgp 65001

bgp router-id 1.1.1.1

neighbor 203.0.113.2 remote-as 65002 ! eBGP - different AS

```
neighbor 10.0.0.2 remote-as 65001      ! iBGP - same AS, requires full-mesh or  
route-reflectors
```

```
network 192.168.1.0 mask 255.255.255.0
```

```
neighbor 203.0.113.2 update-source loopback0
```

```
neighbor 203.0.113.2 password Cisco123
```

BGP Best-Path Selection Order (simplified, memorize this order):

1. Highest Weight (Cisco-local only)
2. Highest Local Preference
3. Locally originated route preferred
4. Shortest AS-Path
5. Lowest Origin type (IGP < EGP < Incomplete)
6. Lowest MED (Multi-Exit Discriminator)
7. eBGP over iBGP
8. Lowest IGP metric to next hop
9. Oldest route (most stable)
10. Lowest Router ID
11. Lowest neighbor IP

Verify: `show ip bgp summary`, `show ip bgp`, `show ip bgp neighbors`.

8.5 Route Redistribution (mixing protocols)

```
router ospf 1
```

```
redistribute eigrp 100 subnets metric 50 metric-type 1
```

```
router eigrp 100
```

```
redistribute ospf 1 metric 10000 100 255 1 1500
```

Always watch for **routing loops** when redistributing bidirectionally between two protocols at multiple points — use route-maps + tagging to prevent re-injection.

9. Access Control Lists (ACLs)

9.1 Standard ACLs (filter by source IP only, numbers 1–99, 1300–1999)

```
access-list 10 permit 192.168.1.0 0.0.0.255
```

```
access-list 10 deny any
```

```
interface gi0/1
```

```
ip access-group 10 out
```

Rule: place standard ACLs **as close to the destination as possible** (since they can't filter by destination, applying near source could block unintended traffic). Wildcard mask = inverse of subnet mask (0=must match, 1=don't care). **0.0.0.255** = match first 3 octets exactly.

9.2 Extended ACLs (filter by source, destination, protocol, port — numbers 100–199, 2000–2699)

```
access-list 110 permit tcp 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255 eq 80
```

```
access-list 110 permit tcp any any established    ! allow return traffic for existing TCP sessions
```

```
access-list 110 deny ip any any log              ! implicit deny is always last — 'log' shows hits
```

```
interface gi0/0
```

```
ip access-group 110 in
```

Rule: place extended ACLs **as close to the source as possible** to drop unwanted traffic early.

9.3 Named ACLs (easier to edit — supports line numbers, insert/delete)

```
ip access-list extended BLOCK-TELNET
```

```
10 deny tcp any any eq 23
```


20 permit ip any any

Editing: **no 10** removes just line 10, then add a new one with a chosen sequence number, e.g. **15 deny tcp host 10.1.1.1 any eq 23**.

9.4 Key Rules & Best Practices

- ACLs are processed **top-down, first match wins** — order matters.
- Every ACL has an **implicit deny any** at the end (invisible but real).
- One ACL per interface, per direction, per protocol (e.g., one inbound IPv4 ACL max per interface).
- Router-generated traffic (from the router itself) is not filtered by an *inbound* ACL on that router unless via other mechanisms (e.g. Control Plane Policing).
- Verify: **show access-lists**, **show ip interface gi0/0** (shows applied ACLs).

9.5 Time-Based ACLs

time-range WORK-HOURS

periodic weekdays 9:00 to 18:00

access-list 110 permit tcp any any eq 80 time-range WORK-HOURS

9.6 Reflexive & Dynamic ACLs (advanced)

- **Reflexive ACL**: auto-creates temporary return-traffic entries for sessions initiated from inside — stateful-like filtering without a full firewall.
- **Dynamic ("lock-and-key") ACL**: requires Telnet/SSH authentication before temporarily opening additional access.

10. NAT / PAT

10.1 Static NAT (1:1 mapping)

ip nat inside source static 192.168.1.10 203.0.113.10

interface gi0/0

ip nat inside

```
interface gi0/1
```

```
ip nat outside
```

10.2 Dynamic NAT (pool-based, many-to-many, no port translation)

```
access-list 1 permit 192.168.1.0 0.0.0.255
```

```
ip nat pool NATPOOL 203.0.113.10 203.0.113.20 netmask 255.255.255.0
```

```
ip nat inside source list 1 pool NATPOOL
```

10.3 PAT / NAT Overload (many-to-one, most common in real networks/home routers)

```
access-list 1 permit 192.168.1.0 0.0.0.255
```

```
ip nat inside source list 1 interface gi0/1 overload
```

10.4 Verifying & Troubleshooting NAT

```
show ip nat translations
```

```
show ip nat statistics
```

```
clear ip nat translation *
```

```
debug ip nat
```

Common issue: forgetting to mark both `ip nat inside` and `ip nat outside` on the correct interfaces — NAT silently does nothing without both.

10.5 Port Forwarding (Static NAT with port)

```
ip nat inside source static tcp 192.168.1.100 80 203.0.113.10 8080
```

11. DHCP

11.1 DHCP Server on IOS

```
ip dhcp excluded-address 192.168.1.1 192.168.1.10
```

```
ip dhcp pool LAN-POOL
```

```
network 192.168.1.0 255.255.255.0
```

```
default-router 192.168.1.1
```

```
dns-server 8.8.8.8 8.8.4.4
```

```
domain-name lab.local
```

```
lease 7                ! days (default) [hours] [minutes] | lease infinite
```

Verify: `show ip dhcp binding`, `show ip dhcp pool`, `show ip dhcp conflict`.

11.2 DHCP Relay (IP Helper) — for clients on a different subnet than the DHCP server

```
interface vlan 20
```

```
ip helper-address 192.168.1.5
```

`ip helper-address` also forwards other broadcast UDP services (TFTP, DNS, NetBIOS, etc.) by default unless restricted with `ip forward-protocol`.

11.3 DORA Process (DHCP client-server exchange)

Discover (client broadcast) → **O**ffer (server) → **R**quest (client broadcasts chosen offer) → **A**cknowledge (server confirms lease).

12. First Hop Redundancy Protocols (FHRP)

Provide a resilient default gateway for hosts when the primary router fails.

12.1 HSRP (Hot Standby Router Protocol) — Cisco proprietary

```
interface gi0/0
```

```
ip address 192.168.1.2 255.255.255.0
```

```
standby 1 ip 192.168.1.1      ! virtual IP, configured on all group members
```

```
standby 1 priority 110        ! higher wins active role (default 100)
```

```
standby 1 preempt             ! allow a higher-priority router to reclaim Active after recovery
```

```
standby 1 track gi0/1 decrement 20  ! object tracking - lower priority if uplink fails
```

```
standby version 2
```

Virtual MAC: `0000.0C9F.Fxxx` (v1) — one Active, one Standby, others Listen.

12.2 VRRP (Virtual Router Redundancy Protocol) — open standard

```
interface gi0/0
```

```
vrrp 1 ip 192.168.1.1
```

```
vrrp 1 priority 110
```

```
vrrp 1 preempt
```

Terms: Master (active) / Backup (vs HSRP's Active/Standby). Virtual MAC: `0000.5E00.01xx`.

12.3 GLBP (Gateway Load Balancing Protocol) — Cisco proprietary, adds load balancing

All members can actively forward simultaneously (unlike HSRP/VRRP's single active router) via a single virtual IP but multiple virtual MACs (AVG assigns MACs to AVFs).

```
interface gi0/0
```

```
glbp 1 ip 192.168.1.1
```

```
glbp 1 load-balancing round-robin
```

Verify: `show standby brief`, `show vrrp brief`, `show glbp brief`.

13. Layer 2 Security

13.1 Port Security

Limits/controls which MAC addresses can use a switch port.

```
interface fa0/1
```

```
switchport mode access
```

```
switchport port-security
```

```
switchport port-security maximum 2
```

```
switchport port-security mac-address sticky    ! learn dynamically & save to config
```

```
switchport port-security mac-address 0011.2233.4455    ! static entry
```

```
switchport port-security violation shutdown    ! shutdown (default, err-disable) | restrict | protect
```

- **shutdown**: err-disables port, sends SNMP/syslog trap (default and most common).
- **restrict**: drops offending traffic, increments violation counter, logs.
- **protect**: drops offending traffic silently, no logging. Verify: `show port-security`, `show port-security interface fa0/1`, `show port-security address`.

13.2 DHCP Snooping

Prevents rogue DHCP servers by trusting only designated uplink ports.

```
ip dhcp snooping
```

```
ip dhcp snooping vlan 10,20
```

```
interface gi0/1
```

```
ip dhcp snooping trust      ! uplink to legitimate DHCP server
```

```
interface fa0/1
```

```
ip dhcp snooping limit rate 10  ! pps, mitigates DoS on untrusted (default) ports
```

Builds the **DHCP snooping binding table** (IP-MAC-VLAN-Port-lease), which feeds Dynamic ARP Inspection and IP Source Guard.

13.3 Dynamic ARP Inspection (DAI)

Prevents ARP spoofing/man-in-the-middle by validating ARP packets against the DHCP snooping binding table.

```
ip arp inspection vlan 10,20
```

```
interface gi0/1
```

```
ip arp inspection trust
```

13.4 IP Source Guard

Prevents IP spoofing by validating source IP/MAC against the DHCP snooping table, per port.

```
interface fa0/1
```

```
ip verify source
```

13.5 Storm Control

Mitigates broadcast/multicast/unicast storms.

```
interface fa0/1
```

```
storm-control broadcast level 50.00
```

```
storm-control action shutdown
```

13.6 VLAN Hopping Mitigations

- Disable DTP (`switchport nonegotiate`), set unused ports to a non-default "parking lot" VLAN and `shutdown` them, never leave native VLAN as VLAN 1, disable auto-trunking, prune unused VLANs off trunks.
-

14. Device Security & AAA

14.1 Password Best Practices

`enable secret Cisco@123` ! preferred (hashed) over 'enable password' (reversible)

`service password-encryption` ! type 7 - weak, but obscures from casual shoulder-surfing

`username admin secret Str0ng@1`

Modern IOS supports stronger hashing: `enable algorithm-type scrypt secret <pw> / username admin algorithm-type scrypt secret <pw>`.

14.2 AAA (Authentication, Authorization, Accounting)

`aaa new-model`

`aaa authentication login default local` ! fallback method list

`aaa authentication login VTY-LOGIN group tacacs+ local`

`aaa authorization exec default group tacacs+ local`

`aaa accounting exec default start-stop group tacacs+`

`tacacs server TAC1`

`address ipv4 10.1.1.5`

`key Cisco123`

`line vty 0 4`

login authentication VTY-LOGIN

RADIUS alternative:

radius server RAD1

address ipv4 10.1.1.6 auth-port 1812 acct-port 1813

key Cisco123

aaa authentication login default group radius local

TACACS+ (TCP 49, encrypts entire packet, separates AAA) vs RADIUS (UDP 1812/1813, encrypts only password, combines authN+authZ) — TACACS+ preferred for device administration, RADIUS common for network access (802.1X, VPN).

14.3 802.1X Port-Based Authentication

aaa new-model

dot1x system-auth-control

interface fa0/1

authentication port-control auto

dot1x pae authenticator

14.4 Control Plane Policing (CoPP) — protects the router's CPU from attack traffic (advanced)

class-map match-all CRITICAL

match access-group 150

policy-map COPP-POLICY

class CRITICAL

police 8000 conform-action transmit exceed-action drop

control-plane

service-policy input COPP-POLICY

15. WAN Technologies

15.1 PPP (Point-to-Point Protocol)

```
interface serial0/0/0
```

```
encapsulation ppp
```

```
ppp authentication chap
```

```
ppp chap hostname R1
```

```
ppp chap password Cisco123
```

CHAP (Challenge Handshake, 3-way, password never sent in clear) preferred over PAP (2-way, plaintext). PPP supports **multilink (MLPPP)** to bundle multiple serial links: `interface multilink1`.

15.2 HDLC

Cisco's default serial encapsulation (`encapsulation hdlc`) — simpler, no authentication, only Cisco-to-Cisco compatible in its proprietary form.

15.3 Frame Relay (legacy WAN, still tested/asked about)

```
interface serial0/0/0
```

```
encapsulation frame-relay
```

```
frame-relay lmi-type ansi
```

```
interface serial0/0/0.100 point-to-point
```

```
ip address 10.0.0.1 255.255.255.252
```

```
frame-relay interface-dlci 100
```

Key terms: **DLCI** (Data Link Connection Identifier, locally significant), **LMI** (Local Management Interface — signaling), **PVC** (Permanent Virtual Circuit).

15.4 GRE Tunnels

Encapsulates a wide variety of protocols across an IP network — no encryption by itself.

```
interface tunnel0
```

```
ip address 172.16.0.1 255.255.255.252
```

```
tunnel source gi0/0
```

```
tunnel destination 203.0.113.5
```

```
tunnel mode gre ip          ! default mode, can be omitted
```

15.5 Site-to-Site IPsec VPN (Crypto Map method, classic)

```
crypto isakmp policy 10
```

```
encryption aes 256
```

```
hash sha256
```

```
authentication pre-share
```

```
group 14
```

```
crypto isakmp key Cisco123 address 203.0.113.5
```

```
crypto ipsec transform-set TSET esp-aes 256 esp-sha256-hmac
```

```
access-list 100 permit ip 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255
```

```
crypto map CMAP 10 ipsec-isakmp
```

```
set peer 203.0.113.5
```

```
set transform-set TSET
```

```
match address 100
```

```
interface gi0/1
```

```
crypto map CMAP
```

GRE-over-IPsec is common when you need routing protocols (OSPF/EIGRP) across the VPN, since plain IPsec (tunnel mode) can't carry multicast/broadcast routing hellos.

15.6 DMVPN (advanced overlay, brief overview)

Combines **mGRE** (multipoint GRE) + **NHRP** (Next Hop Resolution Protocol) + IPsec to build a dynamic hub-and-spoke or spoke-to-spoke VPN mesh without manually configuring every tunnel.

16. QoS Fundamentals

16.1 Why QoS

Manages congestion, delay, jitter, and packet loss for latency-sensitive traffic (voice, video) by classifying, marking, queuing, and policing/shaping traffic.

16.2 Classification & Marking

```
class-map match-all VOICE
```

```
match dscp ef
```

```
policy-map MARK-TRAFFIC
```

```
class VOICE
```

```
set dscp ef
```

```
interface gi0/0
```

```
service-policy input MARK-TRAFFIC
```

Common markings: **DSCP EF** (Expedited Forwarding, voice), **AF** classes (Assured Forwarding, data), **CS** (Class Selector, legacy IP Precedence-compatible).

16.3 Queuing (output congestion management)

policy-map QOS-OUT

class VOICE

priority percent 20 ! LLQ - Low Latency Queuing, strict priority

class DATA

bandwidth percent 50

class class-default

fair-queue

interface gi0/0

service-policy output QOS-OUT

16.4 Policing vs Shaping

- **Policing**: drops (or re-marks) traffic exceeding a rate immediately — no buffering, used for enforcing SLAs on ingress.
- **Shaping**: buffers/delays excess traffic to smooth bursts to a target rate — used on egress, especially toward slower WAN links.

policy-map POLICE-HTTP

class HTTP

police 1000000 conform-action transmit exceed-action drop

policy-map SHAPE-WAN

class class-default

shape average 512000

16.5 Trust Boundaries

Configure switches to trust or override markings from end-devices to prevent users from self-prioritizing traffic:

```
interface fa0/1
```

```
mls qos trust device cisco-phone
```

```
mls qos trust cos
```

17. IPv6

17.1 Address Format

128-bit, 8 groups of 4 hex digits, `::` compresses one run of consecutive zero groups (only once per address). Example: `2001:0db8:0000:0000:0000:0000:0000:0001` → `2001:db8::1`.

17.2 Address Types

- **Unicast Global (GUA):** `2000::/3` — routable on the Internet.
- **Link-Local (LLA):** `fe80::/10` — auto-assigned, mandatory on every interface, never routed.
- **Unique Local (ULA):** `fc00::/7` (typically `fd00::/8`) — private, like RFC1918 for IPv6.
- **Multicast:** `ff00::/8`. No broadcast in IPv6 — replaced by multicast (e.g., `ff02::1` = all-nodes, `ff02::2` = all-routers).
- **Anycast:** same address assigned to multiple devices, nearest one responds.

17.3 Enabling & Configuring IPv6

`ipv6 unicast-routing` ! global, required on routers to actually route IPv6

```
interface gi0/0
```

```
ipv6 address 2001:db8:1::1/64
```

```
ipv6 address fe80::1 link-local
```

ipv6 enable ! generates link-local only, no GUA

ipv6 address 2001:db8:1::/64 eui-64 ! auto-generates host portion from MAC (modified EUI-64)

17.4 Static & Dynamic IPv6 Routing

ipv6 route 2001:db8:2::/64 2001:db8:1::2

ipv6 route ::/0 2001:db8:1::1

router ospfv3 1

router-id 1.1.1.1

!

interface gi0/0

ospfv6 1 area 0

router eigrp NAME

address-family ipv6 unicast autonomous-system 100

eigrp router-id 1.1.1.1

interface gi0/0

ipv6 eigrp 100

17.5 SLAAC & DHCPv6

interface gi0/0

ipv6 nd other-config-flag ! tells hosts to also query DHCPv6 for DNS etc (stateless)

ipv6 dhcp pool DHCPV6POOL

dns-server 2001:4860:4860::8888

domain-name lab.local

interface gi0/0

ipv6 dhcp server DHCPV6POOL

Verify: `show ipv6 interface brief`, `show ipv6 route`, `show ipv6 neighbors`
(equivalent of ARP, called **Neighbor Discovery Protocol / NDP**).

18. IOS File & Image Management

18.1 File Systems

`show file systems`

`dir flash:`

`dir nvram:`

18.2 Backing Up / Upgrading IOS Image

`copy flash: tftp:` ! backup current image

`copy tftp: flash:` ! upload a new image

`verify /md5 flash:new-image.bin` ! confirm integrity

`boot system flash:new-image.bin` ! tell IOS which image to boot next

`show boot`

`reload`

18.3 Boot System Order & Config Register

`config-register 0x2102` ! normal boot (default), reads startup-config

`config-register 0x2142` ! ignores startup-config (used in password recovery)

`show version` ! shows current config register value at the bottom

18.4 Cisco IOS File Naming Convention (example)

`c2900-universalk9-mz.SPA.155-3.M4.bin` → platform (c2900), feature set (universalk9), run-from-RAM/zip (mz), signed package (SPA), version (15.5(3)M4).

19. Password Recovery & Licensing

19.1 Password Recovery (Generic Procedure)

1. Connect via console cable.
2. Power cycle the device; send a `break` signal during boot (within ~60 sec) to enter **ROMMON**.
3. `confreg 0x2142` (skip startup-config on boot) then `reset` / power-cycle.
4. Device boots with no startup-config → enter `enable` (no password).
5. `copy startup-config running-config` to restore the config into RAM.
6. Change passwords: `enable secret NewPass`, update line/vty passwords.
7. `config-register 0x2102` to restore normal boot behavior.
8. `copy running-config startup-config, reload`.

19.2 Licensing (modern IOS/IOS-XE — Smart Licensing)

`show license summary`

`show license status`

`license smart register idtoken <token>`

`license boot level network-advantage add-on dna-advantage` ! (platform-dependent syntax)

Older platforms used `license install <file>` (right-to-use / permanent licenses via activation keys).

20. Monitoring, Logging & Troubleshooting

20.1 Logging

logging buffered 16384

logging console

logging trap informational

logging host 192.168.1.100 ! syslog server

service timestamps log datetime msec

show logging

Syslog severity levels (0=most severe): 0 Emergency, 1 Alert, 2 Critical, 3 Error, 4 Warning, 5 Notice, 6 Informational, 7 Debugging.

20.2 SNMP

snmp-server community public RO

snmp-server community private RW

snmp-server location DC1-Rack4

snmp-server contact netops@company.com

snmp-server host 192.168.1.100 version 2c public

snmp-server enable traps

Prefer **SNMPv3** for production (adds authentication + encryption):

snmp-server group ADMINGRP v3 priv

snmp-server user netadmin ADMINGRP v3 auth sha AuthPass123 priv aes 128 PrivPass123

20.3 Diagnostic Commands

ping 8.8.8.8

ping 8.8.8.8 source loopback0

extended ping (interactively prompts for options: repeat count, size, source, etc.)

traceroute 8.8.8.8

show ip route <destination>

show cdp neighbors detail

show interfaces gi0/0 | include (line protocol|packets)

show ip interface brief | exclude unassigned

debug ip icmp

debug ip routing

debug ip ospf adjacency

undebug all ! (or "u all") — stop ALL debugging, do this fast in production

terminal monitor ! shows debug/log output on vty sessions (not just console)

Debugging is CPU-intensive — never leave **debug all** running on a production device.

20.4 The "OSI Layered" Troubleshooting Method

1. Physical: cabling, **show interfaces** (up/up?), SFP/light status.
2. Data Link: VLAN assignment, trunk config, STP state, encapsulation match.
3. Network: IP addressing, subnet mask, routing table, ACLs blocking.
4. Transport: port blocked by firewall/ACL, NAT translation issue.
5. Application: DNS resolution, service actually running. Work bottom-up (or top-down / divide-and-conquer) systematically rather than guessing.

20.5 IP SLA (advanced monitoring)

ip sla 1

```
icmp-echo 8.8.8.8 source-interface gi0/0
```

```
frequency 30
```

```
ip sla schedule 1 life forever start-time now
```

```
track 1 ip sla 1 reachability
```

```
ip route 0.0.0.0 0.0.0.0 203.0.113.1 track 1
```

Used to conditionally install/remove a route based on real reachability testing (common with FHRP + floating statics for automatic failover).

21. Advanced Routing: Redistribution, Route-Maps, VRF-Lite

21.1 Route-Maps (the Swiss-army-knife of policy control)

Used for redistribution filtering, PBR (Policy-Based Routing), BGP attribute manipulation, NAT route filtering, etc.

```
ip prefix-list PL-10 seq 5 permit 192.168.10.0/24
```

```
route-map REDIST-FILTER permit 10
```

```
match ip address prefix-list PL-10
```

```
set metric 20
```

```
route-map REDIST-FILTER deny 20
```

```
!
```

```
router ospf 1
```

```
redistribute eigrp 100 subnets route-map REDIST-FILTER
```

Route-maps are processed top-down like ACLs, first-match, implicit deny at the end (be careful — an empty/unmatched route-map denies everything by default).

21.2 Policy-Based Routing (PBR) — route based on criteria other than destination IP

```
access-list 101 permit ip 192.168.1.0 0.0.0.255 any
```

```
route-map PBR-MAP permit 10
```

```
match ip address 101
```

```
set ip next-hop 10.0.0.2
```

```
interface gi0/0
```

```
ip policy route-map PBR-MAP
```

21.3 VRF-Lite (Virtual Routing and Forwarding)

Creates multiple independent routing tables on one device — used for multi-tenancy without MPLS.

```
ip vrf CUSTOMER_A
```

```
rd 65000:1
```

```
interface gi0/0
```

```
ip vrf forwarding CUSTOMER_A
```

```
ip address 192.168.1.1 255.255.255.0
```

```
ip route vrf CUSTOMER_A 0.0.0.0 0.0.0.0 192.168.1.254
```

```
show ip route vrf CUSTOMER_A
```

```
ping vrf CUSTOMER_A 192.168.1.254
```

(Modern IOS-XE syntax uses **vrf definition** + **address-family ipv4** instead of the legacy **ip vrf**.)

21.4 MPLS & VXLAN (conceptual overview — enterprise/SP scale)

- **MPLS**: labels packets at ingress (LER) for fast label-switched forwarding through the core (LSR) without a full IP lookup at every hop; underpins L3VPN services.
 - **VXLAN**: encapsulates L2 frames in UDP (port 4789) to extend broadcast domains across L3 (data-center overlay, used heavily with EVPN control-plane in modern fabrics like Cisco ACI).
-

22. Automation: Netmiko, EEM, and Modern IOS-XE

22.1 Embedded Event Manager (EEM) — reactive automation directly on the device

event manager applet INTERFACE-DOWN

event syslog pattern "LINEPROTO-5-UPDOWN.*GigabitEthernet0/1.*down"

action 1.0 syslog msg "ALERT: Gi0/1 went down!"

action 2.0 cli command "enable"

action 3.0 cli command "interface gi0/1"

action 4.0 cli command "shutdown"

action 5.0 cli command "no shutdown"

22.2 Netmiko (Python-based external automation — industry standard for network engineers)

```
from netmiko import ConnectHandler
```

```
device = {  
    "device_type": "cisco_ios",
```

```
"host": "192.168.1.1",

"username": "admin",

"password": "Cisco123",

"secret": "Cisco123",

}

conn = ConnectHandler(**device)

conn.enable()

output = conn.send_command("show ip interface brief")

print(output)

config_commands = ["interface loopback99", "ip address 9.9.9.9 255.255.255.255"]

conn.send_config_set(config_commands)

conn.save_config()

conn.disconnect()
```

22.3 RESTCONF / NETCONF (IOS-XE, model-driven programmability)

show restconf

restconf

netconf-yang

Modern network automation increasingly favors **YANG models** over screen-scraping CLI output — worth learning Ansible's `ios_*` modules and `nccclient/pyang` once comfortable with the CLI basics above.

23. Appendix: Cheat Sheets

23.1 Subnetting Quick Reference

CIDR	Mask	Hosts	Block Size
/24	255.255.255.0	254	256
/25	255.255.255.128	126	128
/26	255.255.255.192	62	64
/27	255.255.255.224	30	32
/28	255.255.255.240	14	16
/29	255.255.255.248	6	8
/30	255.255.255.252	2	4
/31	255.255.255.254	2 (RFC3021, P2P only)	2
/32	255.255.255.255	1 (host route)	1

23.2 Common Well-Known Ports

Port	Protocol	Service
20/21	TCP	FTP (data/control)
22	TCP	SSH
23	TCP	Telnet
25	TCP	SMTP
53	TCP/UDP	DNS
67/68	UDP	DHCP (server/client)
69	UDP	TFTP
80	TCP	HTTP

Port	Protocol	Service
110	TCP	POP3
123	UDP	NTP
143	TCP	IMAP
161/162	UDP	SNMP / SNMP Trap
179	TCP	BGP
443	TCP	HTTPS
445	TCP	SMB
514	UDP	Syslog
3389	TCP	RDP

23.3 Command Reference by Task

Task	Command
Enter privileged mode	<code>enable</code>
Enter global config	<code>configure terminal</code>
Set hostname	<code>hostname <name></code>
Assign IP to interface	<code>ip address <ip> <mask></code>
Bring interface up	<code>no shutdown</code>
Save config	<code>copy running-config startup-config / wr</code>
View routing table	<code>show ip route</code>
View interfaces summary	<code>show ip interface brief</code>
Create VLAN	<code>vlan <id> / name <name></code>
Assign access port	<code>switchport mode access / switchport access vlan <id></code>

Task	Command
Configure trunk	<code>switchport mode trunk</code>
Static route	<code>ip route <net> <mask> <next-hop></code>
Enable OSPF	<code>router ospf <id> / network <net> <wildcard> area <area></code>
Enable EIGRP	<code>router eigrp <as> / network <net> <wildcard></code>
Standard ACL	<code>access-list <1-99> permit/deny <src> <wildcard></code>
Extended ACL	<code>access-list <100-199> permit/deny <proto> <src> <wc> <dst> <wc> eq <port></code>
PAT/NAT overload	<code>ip nat inside source list <acl> interface <int> overload</code>
DHCP pool	<code>ip dhcp pool <name> / network / default-router</code>
HSRP VIP	<code>standby <grp> ip <vip></code>
Port security	<code>switchport port-security</code>
SSH setup	<code>crypto key generate rsa + ip ssh version 2</code>

23.4 Suggested Mastery Path

1. Master CLI navigation, basic device config, interfaces, and subnetting cold — this is non-negotiable foundation.
2. Build small GNS3/EVE-NG/Packet Tracer labs for VLANs + trunking + inter-VLAN routing until muscle memory forms.
3. Layer in STP, EtherChannel, and FHRP on redundant topologies.
4. Get comfortable with static routing, then OSPF single-area, then multi-area, then EIGRP, then BGP basics.
5. Add ACLs, NAT, and DHCP to a working topology — simulate real branch/campus scenarios.
6. Layer 2 & device security (port-security, DHCP snooping, AAA, SSH hardening).
7. WAN technologies + VPNs, then IPv6 parallel-track everything you did in IPv4.

8. Finish with QoS, IOS file/image management, password recovery drills, and automation (Netmiko/EEM) to round out real-world operational skill.

Practice discipline: for every topic above, don't just read the commands — build the topology, break it on purpose, and use the `show/debug` commands to diagnose and fix it. That debugging loop is what actually builds CCNA/CCNP-level mastery.